

# CARMApy: A User-Friendly Open-Source Cloud Microphysics Code in Python

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## Software

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## Summary

Clouds are poorly understood yet incredibly important for understanding our observations of exoplanets. The complexity of clouds is due to the fact that macroscopic observables of clouds are dictated by the properties, and in particular size-distributions, of the tiny particles that make up the cloud. These properties are in turn determined by the physics that happens at the molecular scale. The carmapy package allows for the easy simulation of these microscopic processes and interpretation of these results for comparisons with observable properties.

## Statement of need

CARMApy sits towards the high complexity end of the efficiency-complexity trade-off space. As such, it is best suited for tasks where getting an accurate result or a result with minimal physical assumptions is desired over getting a result fast. CARMApy is best suited for tasks such as:

- Forward modeling the aerosol properties of individual exoplanets
- Post-processing General Circulation Model (GCM) outputs to include more accurate cloud treatments
- Running a grid of models to understand demographic trends in exoplanets and/or brown dwarfs
- Modeling morning/evening limb asymmetries caused by cloud formation and transport
- Validating the result of a retrieval with a detailed forward cloud model
- Understanding which microphysical processes are important to cloud formation in different atmospheric regimes

CARMApy builds upon the previous papers that have used the underlying CARMA model as applied to exoplanets (e.g. [Gao & Benneke, 2018](#); [Powell et al., 2018](#); [Powell & Zhang, 2024](#)). This work generalizes the work done by these papers while also open-sourcing and documenting the code. The code is designed such that it can be used by non-experts in microphysical cloud modeling, such as a motivated undergraduate student or an observer validating a retrieval model.

## State of the field

The community uses a variety of tools and techniques across the efficiency-complexity trade-off spectrum to model clouds on exoplanets. On the extreme efficiency end of the scale, there are techniques such as gray cloud models that assume constant opacity across all wavelengths.

39 These models are thus incredibly fast to run and are often incorporated into retrieval models  
 40 (e.g. [Xue et al., 2024](#)). Slightly more complex on this scale are equilibrium cloud condensation  
 41 models ([Ackerman & Marley, 2001](#)). These models incorporate some physics, assuming that  
 42 clouds only form in regions of the atmosphere which are supersaturated.  
 43 These models require a tunable parameter,  $f_{\text{sed}}$ , that accounts for the removal of particles from  
 44 the atmosphere via sedimentation. This framework has been incorporated into open source  
 45 python models such as Virga ([Batalha et al., 2025](#)). A bit more complex, and thus slower to  
 46 run, than those models are the moment-method kinetic cloud models which can incorporate  
 47 detailed microphysics and chemistry calculations, but require a prescribed functional form for  
 48 the particle size distributions. Open-source versions of these models such as mini-cloud (e.g.  
 49 [Lee & Ohno, 2025](#)) exist, but those models are not designed and packaged for widescale use.

50 One more step up this complexity ladder, which is where CARMApy sits, are bin-scheme  
 51 microphysical models. These models are able to incorporate detailed microphysical modeling  
 52 while resolving arbitrary particle size distributions. The Community Aerosol and Radiation  
 53 Model for Atmospheres, CARMA ([Bardeen et al., 2008](#); [Toon et al., 1988](#); [Turco et al., 1979](#)),  
 54 originally designed for Earth clouds, is the most widely used of this class of models, and is  
 55 open source. There have been a number of forks of the CARMA code to apply the code to  
 56 non-Earth planets such as PlanetCARMA ([Barth, 2020](#)), which is currently non-public, and  
 57 ExoCAM-CARMA, which only supports hazes and not clouds and while open-source is currently  
 58 minimally documented. CARMApy builds on the ExoCARMA ([Gao et al., 2018](#); [Powell et al., 2018](#))  
 59 version of CARMA which has extended the functionality of CARMA to include condensates and  
 60 background atmospheres not otherwise modelable with CARMA. Until the publication of this  
 61 work, however, the source code of ExoCARMA has not been public. Additionally, all of these  
 62 codes are written in Fortran over a almost half-century long development history, leading to  
 63 code that only a handful of expert users are able to effectively use. CARMApy is designed to  
 64 make the scientific power of the CARMA family of codes accessible to be used by the general  
 65 exoplanet community.

## 66 Software design

67 The fundamental design philosophy of CARMApy is to expose the core functionality of CARMA  
 68 to the user in an easy to understand manner. This means that we designed CARMApy to be  
 69 opinionated, having default values and implementations for the common ways of using the code  
 70 (e.g. automatically allowing particles to grow or evaporate upon specification of a nucleation  
 71 pathway that forms these particles). While these defaults are strong, and often will occur  
 72 without requiring the user explicitly invoke them, we allow the user to override these defaults  
 73 to maintain feature parity with the underlying CARMA code.

74 Due to the reliance of CARMA on custom types and on `selected_real_kind` for double precision  
 75 calculations, we found it difficult to directly bind the CARMA types and functions from fortran  
 76 to python directly. Instead, upon installation CARMApy compiles our version CARMA into an  
 77 executable that reads all relevant input and configuration data from files which the python layer  
 78 writes. The `Carma.run()` function which runs the simulation will write these input files and run  
 79 the executable. Once the simulation finishes running, the user can call `Carma.read_results()`  
 80 to load the results written to file by the CARMA simulation into a python `Results` object for  
 81 easy analysis by the user. This translation between layers happens once in each direction per  
 82 run and thus does not significantly contribute to computational overhead.

83 A complete description of the physical processes and numerical methods is provided in the  
 84 associated paper ([Cukier et al., 2026](#))

## Research impact statement

CARMApy has a high potential for research impact. The ExoCARMA code which CARMApy upgrades and wraps has already been used in a large number of papers (Gao & Benneke, 2018; Mang et al., 2024; e.g., Powell et al., 2018; Rooney et al., 2022; Wong et al., 2020), although the vast majority of these papers have relied on a small handful of coauthors to run the CARMA model. By bringing this code to python, along with providing detailed documentation and tutorials, the number of researchers who would be able to run CARMApy is significantly greater.

CARMApy is already being used in a number of in prep papers [Cukier et al. in prep; Kennedy et al. in prep; Steinrueck et al. in prep; Samra et al. in prep] with the users spanning career stages from undergraduate to post doc, and these users providing feedback on documentation, reporting bugs, and suggesting new features.

We additionally have performed benchmarking our code to the versions of ExoCARMA previously used in publications. For a sample brown dwarf case run on a M4 Macbook Pro we find that our code runs 3.8 times faster than previous versions when taking advantage of the newly implemented multithreading capabilities of CARMApy and 1.9 times faster than the previous versions while running in single threaded mode. These speedups occur while reproducing the observables, in this case a thermal emission spectrum, to 3 parts in 10,000, a number which is consistent with floating point error caused by our optimizations.

## AI Usage Disclosure

AI tools (Sonnet 4.6, Opus 4.7, Opus 4.8) were used to help generate packaging code, generate the test suite, refactor the codebase, tweak the documentation, optimize the codebase, and implement helper functions. All AI generated code has been read through and checked by a human, with extra scrutiny given to any code that touches the core simulation logic. Additionally, manual benchmarking was performed to ensure that any AI generated code did not unexpectedly change the physics of the simulation.

The subset of the above AI tools were used for this paper to format metadata and citations, which were then checked manually, and provide suggestions on how to improve the prose, which were accepted, rejected, and implemented manually.

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